

HES - MDM Integration

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Role	Name	Decision	Remarks

Revision History

Version	Date	Author	Notes
V1.0	2024-08-14	Jaswant Singh Choudhary	API definition for all features supported in HES SMA.
V1.1	2024-08-15	Jaswant Singh Choudhary	1. TOU SET API Structure updated for Linkwell 3 phase, HPL 3 Phase, Genus 1 and 3 phase Meters in section 6.13 2. Network PING API added new as section 12. 3. Section 3 has been added to defined structure of Meter Job API. 4. Updated section 5.2, 6, 6.12.
V1.2	2024-10-17	Jaswant Singh Choudhary	1. TOU SET API added for AEW MM for 1p and 3p in section 6.13
V1.3	2025-01-16	Jaswant Singh Choudhary	1. LRCF API support added newly as section 6.20 with updated structure
V1.4	2026-04-01	Jaswant Singh Choudhary	1. Section 3 is updated to add details for giving job name. 2. Updated Job responses in all sections.
V1.5	2026-04-20	Jaswant Singh Choudhary	1. Section 4.1 and 4.2 are updated with instructions to do ODR for Instantaneous profile.
V1.6	2026-09-10	Jaswant Singh Choudhary	1. New section added as 6.19 for meter firmware upgrade over the air (FOTA) 2. Section 6.12 updated for Schneider meter.

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1 Overview

HES-MDM Integration can be done three ways as follows:

- A. **Integration through APIs** – This type of Integration must be followed as usual practice through which majority of data/actions will flow in form of APIs.
- The Omnimesh SMA API is provided using RESTful web services that can be accessed via either unsecured HTTP (only recommended for secure private networks) or secure HTTPS transport.
 - The data interchange format for all resources is JSON.
 - This API document includes a detailed definition of each available method, including its inputs.
 - In addition to this process workflow examples are given for some specific use cases.
 - It should be noted that some of the API responses may contain additional fields not defined in this document. These should be considered legacy or development support features and should not be relied upon, they may be removed on production deployments or in future releases.
 - If any specific additional API help required, MDM should reach out to respective Project Manager.
 - Access to all SMA web services is protected by a username and password (basic auth mechanism) access control built into the HTTP server. The web services are available on the following URL:
<https://servername/sma/ws>
 - Overall end to end integration between HES and MDMS consists of two types of APIs Sync and Async.
 - For every Async API run, MDM to provide the callback URL (without authentication) and timeout time in input JSON request as mandatory parameter. Callback URL for sending the response to MDMS and timeout parameter so that every request initiated from MDMS run for defined period.
 - There are two types of call back defined
 1. JOB_METER_CALLBACK
 2. JOB_CALLBACK
 - JOB_METER_CALLBACK should be the default callback type.
 - JOB_CALLBACK is useful in two cases only:
 1. If the MDMS needs to know when the overall job has finished (and can't be bothered to track this itself...)
 2. If there is a lot of on-demand reading data which is too big for JOB_METER_CALLBACK; use JOB_CALLBACK to find out when the job is done, then use the APIs to read one page of the response at a time.

- API response to MDM in both callback type will vary based on callback type defined in API request and sample structure for both cases have been defined in section 4.1 and wrap up for call back response will remain uniform across all use cases.
 - **DomainName will be HES URL in APIs URL.
 - For all major Meter Operations (GET/SET), the Meter Job API will be used; the API is configured to support **one meter per job only**, ensuring better SLA adherence and simplified handling of individual meter requests, including cancellation.
 - With the use of Meter Job API, with single API MDMs can create a group, create a Job, start a job and with help of call back functionality in same Meter Job API, MDMs will get response as well. Hence no need to use multiple APIs.
- B. **Integration through Messaging Queues** – This Type of Integration uses messaging queue-based system, which is basically used to push meter data flowing from meter to MDM via HES into MDM Defined Queues. The data will be sent in JSON format in MDM Queues.
- C. **File based Integration** – This type of Integration is suggested in case of system network down or connectivity failure scenarios, where .csv/.json files can be provided by HES and MDM must be able consume those files and store data as it is happening in real time integration through APIs.

2 OBIS Code for Profile(s)

Metering Profile(s)	Obis Code
Nameplate Profile	0.0.94.91.10.255
Instantaneous Profile	1.0.94.91.0.255
Block Load Profile	1.0.99.1.0.255
Daily Load Profile	1.0.99.2.0.255
Billing Profile	1.0.98.1.0.255
Event-Current	0.0.99.98.1.255
Event-Voltage	0.0.99.98.0.255
Event-Power	0.0.99.98.2.255
Event-Transaction	0.0.99.98.3.255
Event-Others	0.0.99.98.4.255
Event- non-rollover	0.0.99.98.5.255
Event-Control	0.0.99.98.6.255

3 Meter Job API Structural definition

JSON Tag	Definition	Data Type
jobConfiguration	This tag works as a wrap to cover the configuration for job being initiated.	-
commands	This tag defines exact details and configuration for a job.	-
type	This tag defines type of command being executed, as mentioned in section 3.1.	String Predefined value
jobType	This tag defines type of Job, as mentioned in section 3.2.	String Predefined value
jobName	This tag is unique every time with respect to each request. Job Name shouldn't have "." character, "_" is suggested to be used if needed.	String
meters	This tag is used to define meter on which job is being initiated.	Array String
startJob ()	This tag is used to start the job immediately.	String Predefined value 1. true
timeoutTime	This tag is used to define timeout time for a job, so that job runs till defined time, not till infinity time.	Date time stamp (yyyy-mm-ddTHH:MM:SS+5:30)
callback (Optional)	This tag is used as wrap up to define callback type and call back URL	-
callbackOn (Optional)	This tag is used to defined type of call back as in section 3.5.	String Predefined Value
callbackUrl (Optional)	This tag is used to define callback URL, mostly it will be defined by MDMs where they are expecting response to be sent.	String

3.1 jobConfiguration

Structure defining a job, including the payload to be sent to meters.

Definition of a meter job, including the payload to be sent to the meters.

Field	Type	Definition
Commands	MeterJobCommand	The list of commands to be performed in the job.

3.2 MeterJobCommand

Structure defining a command job.

This is split into two parts; the common parts defined here, and the **type** specific set of fields defined in [MeterJobCommandType](#).

Field	Type	Default	Definition
Type	MeterJobCommandType	REQUIRED	The type of command. {Refer section 3.3}
Active	Boolean	REQUIRED for GET operations	This has no effect for METER_COMMAND_SET or METER_FIRMWARE_UPGRADE . For METER_COMMAND_GET operations, this controls whether to request the currently active configuration (true) or where supported the pending configuration (False) often referred to as “passive”.
activationTime	Date time stamp (yyyy-mm-ddTHH:MM:SS+5:30)	OPTIONAL	Specify the time of activation of the configuration. For meters that support deferred activation (passive settings) this will update this field and set the relevant activation time. If not specified, or the meter doesn't support deferred activation, the configuration will be immediately applied. For ex: 2024-08-22T13:50:00+05:30

3.3 MeterJobCommandType

Enumerated list defining the type of meter command jobs, represented as JSON strings.

For each type of command specific payload definition is also referenced.

Value	Definition
PROFILE_CAPTURE_PERIOD	Configure the profile capture period.
DEMAND_INTEGRATION_PERIOD	Configure the demand integration period.
BILLING_PERIOD	Configure the billing period.
TARIFF_CALENDAR	Configure the tariff calendar.
LOAD_CURTAILMENT	Configure the load curtailment operation.
PAYMENT	Configure the payment model and state.
MAX_DEMAND_RESET	Reset the maximum demand registers. Note: GET operation not supported.
DISCONNECTOR_CONTROL	Set the state of the disconnecter.
METERING_MODE	Configure metering mode of operation.
PING	Ping meters.
ON_DEMAND_PROFILE	Request sample for meter profile, for a group.

3.4 MeterJobType

Enumerated type defining the type of job, represented as JSON strings.

Value	Definition
METER_COMMAND_SET	This job performs an action or configuration update on a meter.
METER_COMMAND_GET	This job requests the status from the meter.
METER_FIRMWARE_UPGRADE	This job performs a firmware upgrade on the meter.
METER_ACTION	This job performs Ping or OnDemandProfile on a meter.

3.5 MeterJobCallback

Structure defining the job callback information. Provides information about the type of callback(s) to be executed and the URL to be called.

Field	Type	Default	Definition
callbackOn	MeterJobCallbackType	REQUIRED	The list of the callbacks to be executed {Refer section 3.6}
callbackUrl	String	REQUIRED	The URL of the API to be called.

3.6 MeterJobCallbackType

Enumerated type defining the type of callback to be called by a job, represented as JSON string.

Value	Definition
JOB_METER_CALLBACK	Callback will be executed when a meter task of the job completes.
JOB_CALLBACK	Callback will be executed when the job completes.

3.7 MeterJobSummary

Structure defining the summary status of a job. Provides information about the current job configuration and number of meters which have been successfully processed.

Field	Type	Definition
jobName	String	The name of the job.
meterGroup	String	The meter group on which this job is performed.
scheduledTime	Date time stamp (yyyy-mm-ddTHH:MM:SS+5:30)	Time at which the job was scheduled to execute.
timeoutTime	Date time stamp (yyyy-mm-ddTHH:MM:SS+5:30)	Time at which the job will timeout, if not already completed.
createTime	Date time stamp (yyyy-mm-ddTHH:MM:SS+5:30)	Time at which the job was created.
updateTime	Date time stamp	Time at which the job status was last updated.

	(yyyy-mm-ddTHH:MM:SS+5:30)	
status	MeterJobStatus	The current overall status of the job. {Refer section 3.10}
jobType	MeterJobType	The type of job. {Refer section 3.4}
jobConfiguration	MeterJobConfig	The configuration of the job.
totalProcessedDevices	Integer	The number of meters that have been processed within the job.
totalSuccessfulDevices	Integer	The number of meters that have successfully completed.
totalFailedDevices	Integer	The number of meters that have failed to complete. This will only be meters that have reported permanent, non-communications related, failures.

3.8 MeterJobResponse

Structure defining the command response for GET operations.

This is split into two parts; the common parts defined here, and the **type** specific set of fields defined in MeterJobCommandType.

Field	Type	Default	Definition
Type	MeterJobCommandType	REQUIRED	The type of command. {Refer section 3.3}
Active	Boolean	true	Whether the request was for the active or passive configuration setting.

Structure defining the command response for METER_ACTION Ping operations.

Field	Type	Default	Definition
elapsedTime	epoch milliseconds	N/A	Elapsed time of the ping request to the meter in milliseconds

Structure defining the command response for METER_ACTION On Demand profile

operationsField	Type	Definition
samples	MeterSampleOnDemand	The list of retrieved metering samples. {Refer section 4.1}

3.9 MeterJobMeterStatus

Structure defining the status of an individual meter within a job.

Field	Type	Default	Definition
meterId	String	REQUIRED	The ID of the meter the status relates to.
status	MeterJobMeterStatusType	REQUIRED	The status of the job on this meter. {Refer section 3.11}
failureStep	Integer	REQUIRED	If the job has failed, this will indicate which command in the sequence failed.
response	MeterJobResponse	OPTIONAL	For METER_COMMAND_GET and METER_ACTION job types, it will contain the response from the meter. {Refer section 3.8}

3.10 MeterJobStatus

Enumerated type defining the status of a job, represented as JSON strings.

Value	Definition
SCHEDULED	Job has been created but not yet started.
RUNNING	Job is currently active, and meters are being processed.
FINISHED	Job has finished running for all meters. This does not imply all meters have completed successfully, just that no more work is being done.
TIMEOUT_CANCELLED	Job has timed out; some meters may have failed to complete.
CANCELLING	Job is in the process of being cancelled.
CANCELLED	Job has been cancelled at the request of the user.

3.11 MeterJobMeterStatusType

Enumerated type defining the status of a meter within a job, represented as JSON strings.

Value	Definition
RUNNING	Job has not yet completed on the meter.
SUCCESS	Job has completed successfully on the meter.
FAILED	Job has failed to complete on the meter.
TIMEOUT_CANCELLED	Job has timed out before completing on the meter.
DEVICE_UNREACHABLE	Job failed to reach the meter due to a communications failure.

3.12 MeterJobCallbackPayloadResponse

Structure defining the meters status within the job callback payload response.

Field	Type	Default	Definition
meters	MeterJobMeterStatus	REQUIRED	The status information for each meter within a job. It will contain either response for a single meter (when the callback is triggered by a single job meter completion or for all meters (when the callback is triggered by the job completion). {Refer section 3.9}

3.13 MeterJobCallbackPayload

Structure defining the job callback payload sent at the time a job or a meter job is completed, depending on MeterJobCallbackType value defined on input parameters in /meterJob API.

Field	Type	Definition
jobName	String	The name of the job.
callbackType	MeterJobCallbackType	The type of callback being executed. {Refer section 3.6}
JobMeterStatusResponse	MeterJobCallbackPayloadResponse	The status of meters. {Refer section 3.12}

4 On-Demand Read (ODR)

4.1 Using Async – Meter Job API with call back

API URL	DomainName/sma/ws/meterJob
API Method	POST

Field	Type	Default	Definition
formattedProfileObisCode	String	REQUIRED	The OBIS code of the profile object to read.
sampleStartTime	Date time stamp (yyyy-mm-ddTHH:MM:SS+5:30) For Ex.: 2024-08-22T13:50:00+05:30	REQUIRED	The start of the time period to retrieve from the profile.
sampleStopTime	Date time stamp (yyyy-mm-ddTHH:MM:SS+5:30) For Ex.: 2024-08-22T13:50:00+05:30	REQUIRED	The end of the time period to retrieve from the profile.

- Need to change the OBIS code (mentioned in the point #2) according to the requirement for which metering profile need to do On Demand Read.
- In case of Group ODR action, meterJob API should not be run for more than 1k number of meters.
- Difference between Start and Stop time in API request should not be beyond 3 Days.
- For the instantaneous profile, “**sampleStartTime**” shall be set to 15 minutes prior to the current time and “**sampleStopTime**” shall be set to 10 minutes prior to the current time.

Sample Request (Power Event) Meter Job API:

```

-----API Request-----
{
  "jobConfiguration":
  {
    "commands":
    [
      {
        "type": "ON_DEMAND_PROFILE",
        "formattedProfileObisCode": "0.0.99.98.2.255",
        "sampleStartTime": "2024-08-12T00:00:00+05:30",
        "sampleStopTime": "2024-08-12T15:00:00+05:30"
      }
    ],
    "jobType": "METER_ACTION",
    "jobName": "Unique_Job_Name",
    "meters":
    [
      "A6725527"
    ],
    "startJob": true,
    "timeoutTime": "2024-08-12T17:25:00+05:30",
    "callback":
    {
      "callbackOn":
      [
        "JOB_METER_CALLBACK",
        "JOB_CALLBACK"
      ],
      "callbackUrl": "MDMCallBackURL"
    }
  }
}

```

-----Callback Response-----

1. JOB_CALLBACK Response for all meters in single Job



Response_JOB_CALLBACK_ODR.json

2. JOB_METER_CALLBACK Response for each meter



Response_JOB_METER_CALLBACK_A67255

Response code

Code	Definition
201	Request Initiated
401	Unauthorized
403	Forbidden
404	Not Found

4.2 Sync API

API URL	DomainName/sma/ws/onDemandProfile/\$meterID
API Method	POST

Field	Type	Default	Definition
formattedProfileObisCode	String	REQUIRED	The OBIS code of the profile object to read.
sampleStartTime	Date time stamp (yyyy-mm-ddTHH:MM:SS+5:30) For Ex.: 2024-08-22T13:50:00+05:30	REQUIRED	The start of the time period to retrieve from the profile.
sampleStopTime	Date time stamp (yyyy-mm-ddTHH:MM:SS+5:30) For Ex.: 2024-08-22T13:50:00+05:30	REQUIRED	The end of the time period to retrieve from the profile.
persistReadings	Boolean	OPTIONAL	true: Reading will persist. false: Reading will not persist.

- This API triggers ODR for one meter/request and sends back Sync response immediately based on meter connectivity and availability.
- “\$meterID” will be variable parameter, which will define as Meter Serial Number for which need to do ODR.
- With this API approach as well, Group ODR can be performed by sending multiple requests from MDM, which in case at last results into N requests to Meters via HES with N Responses back to MDMS.
- For the instantaneous profile, “sampleStartTime” shall be set to 15 minutes prior to the current time and “sampleStopTime” shall be set to 10 minutes prior to the current time.

Sample Request (Power Event) Sync API:

API Request

```

{
  "formattedProfileObisCode": "0.0.99.98.2.255",
  "cached": false,
  "persistReadings": false,
  "sampleStartTime": "2024-02-13T01:00:00+05:30",
  "sampleStopTime": "2024-02-14T01:00:00+05:30"
}

```

Response


 ODR_event_Response.txt

Response Body

```

{
  "samples": [
    {
      "time": 1718795294,
      "registerValues": [
        {
          "registerObisCode": "AAABAAD/",
          "formattedRegisterObisCode": "0.0.1.0.0.255",
          "formattedValue": "2024-06-19T16:38:14+05:30",
          "attributeId": 2,
          "unit": 0,
          "scalar": 0,
          "registerValue": "CQwH6AYT/xAmDv8BSgA=",
          "description": ""
        },
        {
          "registerObisCode": "AABgCwL/",
          "formattedRegisterObisCode": "0.0.96.11.2.255",
          "formattedValue": "102",
          "attributeId": 2
        }
      ]
    }
  ]
}

```

Response code

Code	Definition
201	Request Initiated
401	Unauthorized
403	Forbidden
404	Not Found

5 Sequence Wise Scheduled Meter Data facilitation to MDM

5.1 Meter Sample API

API URL	DomainName/sma/ws/meterSamples
API Method	POST

Field	Type	Definition
meterSampleId	Integer	The monotonic sequence number, this is unique within the buffer. This value should be considered 64-bits (signed) in size.
sequenceNumber	Integer	Sequence in which the samples are received from the Agent.
deviceId	String	The associated meter identifier.
nodeId	String	The communications node this meter is connected to.
profileObisCode	String	The associated formatted profiles OBIS code.
formattedProfileObisCode	String	The associated formatted profiles OBIS code.
registerValues	MeterRegister	The list of register values.
sampleTime	Date time stamp (yyyy-mm-ddTHH:MM:SS+5:30) For Ex.: 2024-08-22T13:50:00+05:30	The time at which the sample was created.
createTime	Date time stamp (yyyy-mm-ddTHH:MM:SS+5:30) For Ex.: 2024-08-22T13:50:00+05:30	The time at which the sample was received by the SMA.

- This API can be used to pull data based on sample ID and sends back Sync response immediately.
- This API can work as a streaming API by defining parameters like “**count**” and “**startId**”.
- Maximum value for “**count**” parameter will be 10K for single run.
- With the help of this API, all metering profiles inclusive of events data can be fetched from HES.
- There will not be any mechanism in this API to fetch details profile or event wise, it will be totally based on sample ID on which data is stored in HES.

-----API Request-----

```
{
  "count": 2,
  "startId": 1
}
```

Response


 MeterSample.txt

Response Body

```
[
  {
    "meterSampleId": 141934,
    "sequenceNumber": 437,
    "deviceId": "Z00000294",
    "nodeId": "00-1b-c5-0c-60-47-07-61",
    "profileObisCode": "AQBeWwD/",
    "formattedProfileObisCode": "1.0.94.91.0.255",
    "registerValues": [
      {
        "registerObisCode": "AAABAAAD/",
        "formattedRegisterObisCode": "0.0.1.0.0.255",
        "formattedValue": "2024-03-25T15:00:32+05:30",
        "attributeId": 2,
        "unit": 0,
        "scalar": 0,
        "registerValue": "CQwH6AMZAQ8AIP8BSgA=",
        "description": null
      },
      {

```

Response Code

200

Response code

Code	Definition
201	Request Initiated
401	Unauthorized
403	Forbidden
404	Not Found

5.2 Meter Alarms API

API URL	DomainName/sma/ws/meterAlarms
API Method	POST

Field	Type	Definition
meterId	String	The associated meter identifier.
meterAlarmId	Integer	The monotonic sequence number, this is unique within the buffer. This value should be considered 64-bits (signed) in size.
sequenceNumber	Integer	Sequence in which the samples are received from the Agent.

alarmTime	Date time stamp (yyyy-mm-ddTHH:MM:SS+5:30) For Ex.: 2024-08-22T13:50:00+05:30	The time at which the event mask was updated.
alarmActive	MeterAlarmType	A list of alarms that are currently active in the meter. {Refer section 5.2.1}
createTime	Date time stamp (yyyy-mm-ddTHH:MM:SS+5:30) For Ex.: 2024-08-22T13:50:00+05:30	The time at which the alarm is created at HES.

5.2.1 Meter Alarm Type

Value	Definition
METER_PHASE_MISSING	One of the phases is missing.
METER_OVER_VOLTAGE	One of the phases is over voltage.
METER_UNDER_VOLTAGE	One of the phases is under voltage.
METER_VOLT_UNBALANCE	Voltage is unbalanced.
METER_CURRENT_REVERSE	One of the phases has the current reversed.
METER_CURRENT_UNBALANCE	The current is unbalanced.
METER_CURRENT_BYPASS	The current has been bypassed / shorted.
METER_OVER_CURRENT	One of the phases is over current.
METER_LOW_PF	Very low power factor.
METER_EARTH_LEAKAGE	Earth leakage detected.
METER_MAGNETIC_INFLUENCE	Magnetic influence detected (tampering).
METER_NEUTRAL_DISTURBANCE	Neutral disturbance detected (tampering).
METER_COVER_OPEN	The meter cover has been opened (tampering).
METER_CONNECT_DISCONNECT	The meter load has been connected / disconnected.
METER_LAST_GASP	Meter has experienced a power failure.
METER_FIRST_BREATH	Meter has powered on following an outage.
METER_BILLING_INCREMENT	The billing counter has been incremented.
METER_PHASE_OPEN	Something is wrong with the current transformer on any phase.
METER_SWITCH_WELD	The disconnecter relay has failed (fused).
METER_NEUTRAL_MISS	Neutral missing.
METER_OVER_LOAD	Meter over load limit.

- This API can be used to pull Alarms based on sample ID and sends back Sync response immediately.
- This API can work as a streaming API by defining parameters like “**count**” and “**startId**”.
- Maximum value for “**count**” parameter will be 10K for single run.
- With the help of this API, all alarms can be fetched from HES.

Response code

Code	Definition
201	Request Initiated
401	Unauthorized
403	Forbidden
404	Not Found

-----API Request-----

```

{
  "count": 2,
  "startId": 1
}

```

-----Response-----



Meter Alarms.json

```

[
  {
    "meterId": "A5543957",
    "meterAlarmId": 12,
    "sequenceNumber": 5836,
    "alarmTime": "2023-10-06T13:43:06+05:30",
    "alarmActive": [
      "METER_LAST_GASP"
    ],
    "createTime": "2024-06-05T12:45:15+05:30"
  },
  {
    "meterId": "A5543957",
    "meterAlarmId": 13,
    "sequenceNumber": 5850,
    "alarmTime": "2023-10-06T16:46:23+05:30",
    "alarmActive": [
      "METER_FIRST_BREATH"
    ]
  },
]

```

6 Meter Operation

API URL	DomainName/sma/ws/meterJob
API Method	POST

Response code

Code	Definition
201	Request Initiated
400	Bad Request
401	Unauthorized
403	Forbidden
404	Not Found

- All SET Action must be performed after getting approval from designated Authorities.
- To meet SBD service levels and to support cancellation of a DC job before issuing an RC job, APIs should use one meter per job, means action must be performed only on individual Meter in one run.
- All DC or RC jobs should have single meter only in one job.
- For RC jobs, need to set timeout time of 6 hrs and till then HES will keep re-trying until 6 hrs timeout hits the job.
- Those RC jobs which got timeout after 6 hrs. So, these cases will mostly be the non-Comm cases so ticket should be generated for Manual Reconnection and assign to field team for action.
- All DC Jobs are expected to be start by 10 AM and generally DISCOM cutoff window is 4 PM, so all DC jobs should have a timeout time of 6 hrs as 16:00:00 of that day. (These details may vary project to project and DISCOM to DISCOM)
- All DC jobs whose response not received to MDM by 4PM of the day should be cancelled/timeout and then to be retry next working day.
- If any DC job is issued and response not received and in between Recharge is received, then that DC job should be cancelled by MDM and issue a new RC job.
- For Billing Profile SET, only **“dayOfMonth”** can be set from 1 to 28, it is not allowed to set for 29, 30 and 31.
 - *** If meter supports bill date to set as 29, 30 and 31, HES will be SET same on meter.
- For Billing Profile SET, **“billingCycle”** can be SET only in case meter supports respective functionality.
- For TOU SET functionality MDMs must follow below steps
 - Utility raised a request to SET TOU on production field meters to AMISP.
 - AMISP will reach out to MDMs to execute same.
 - MDMs will review request and accordingly will SET TOU, firstly for few test meters of same meter manufacturer and type.
 - MDMs will convey same SET Time of zones to Meter Manufacturer and get written/email approval, that whatever time of zones has been defined are as per request from Utility and will capture billing consumption accordingly.
 - After getting written/email approval from meter vendor or manufacturer, then MDMs need to reach out to Utility to get written/email approval to SET same TOU zones in production field meters in bulk by defining activation time specifically in approval letter.
 - Generally, TOU must be SET at the beginning of month i.e., 1st of any month with activation time as 00:00:00. Activation time must be set as like 00:30:00 or 01:00:00 or 01:30:00, it shouldn't be sent in some odd timings like 00:15:00 or 00:20:00 etc.

- Below is the list for the activity/operation for which need to perform the API run and get the response.

Sr. No.	Operations/Activities	Sr. No.	Operations/Activities
1	Relay Status	11	Billing Profile (SET)
2	Disconnect	12	Activity Calendar (GET)
3	Reconnect	13	Activity Calendar (SET)
4	Ping Meter	14	Load Curtailment (GET)
5	Demand Integration Period (GET)	15	Load Curtailment (SET)
6	Demand Integration Period (SET)	16	Enable & Disable Net Metering
7	Profile Capture Period (GET)	17	Enable & Disable Prepaid
8	Profile Capture Period (SET)	18	Recharge Balance Display on Meter Screen (SET)
9	Max. Demand Reset	19	Last Token Recharge Amount (GET)
10	Billing Profile (GET)	20	LRCF – Local relay connect functionality

6.1 Relay Status

API Request
<pre> { "jobConfiguration": { "commands": [{ "type": "DISCONNECTOR_CONTROL", "active": "True" }], "jobName": "Unique_Job_Name", "jobType": "METER_COMMAND_GET", "meters": ["A6725527"], "startJob": true, "timeoutTime": "2024-06-24T13:05:00+05:30", "callback": { "callbackOn": ["JOB_METER_CALLBACK", "JOB_CALLBACK"], "callbackUrl": "MDMCallBackURL" } } }</pre>
Callback Response



Get Relay Status Job
Call Back Response.js

6.2 Disconnect

-----API Request-----

```
{
  "jobConfiguration":
  {
    "commands":
    [{
      "type": "DISCONNECTOR_CONTROL",
      "controlRequest": "DISCONNECT"
    }]
  },
  "jobName": "Unique_Job_Name",
  "jobType": "METER_COMMAND_SET",
  "meters":
  [
    "ZR1J00006"
  ],
  "startJob": true,
  "timeoutTime": "2024-06-24T13:05:00+05:30",
  "callback":
  {
    "callbackOn":
    [
      "JOB_METER_CALLBACK",
      "JOB_CALLBACK"
    ],
    "callbackUrl": "MDMCallbackURL"
  }
}
```

-----Callback Response-----



Disconnect Call Back
Response.json

6.3 Reconnect

-----API Request-----

```
{
  "jobConfiguration": {
    "commands": [{
      "type": "DISCONNECTOR_CONTROL",
      "controlRequest": "CONNECT"
    }]
  },
  "jobName": "Unique_Job_Name",
  "jobType": "METER_COMMAND_SET",
  "meters": [
    "ZR1J00006"
  ],
  "startJob": true,
  "timeoutTime": "2024-06-24T13:05:00+05:30",
  "callback": {
    "callbackOn": [
      "JOB_METER_CALLBACK",
      "JOB_CALLBACK"
    ],
    "callbackUrl": "MDMCallBackURL"
  }
}
```

-----Callback Response-----



**RECONNECT Call
Back Response.json**

6.4. Ping Meter

-----API Request-----

```
{
  "jobConfiguration": {
    "commands": [{
      "type": "PING"
    }]
  },
  "jobType": "METER_ACTION",
  "jobName": "Unique_Job_Name",
  "meters": [
    "A6725527"
  ],
  "startJob": true,
  "timeoutTime": "2024-06-24T13:55:00+05:30",
  "callback": {
    "callbackOn": [
      "JOB_METER_CALLBACK",
      "JOB_CALLBACK"
    ],
    "callbackUrl": "MDMCallBackURL"
  }
}
```

-----Callback Response-----



**Ping Call Back
Response.json**

6.4 Demand Integration Period (GET)

Field	Type	Default	Definition
demandPeriod	Period	REQUIRED	The required demand integration period in seconds. Period: defined in milli-seconds (ms)

-----API Request-----

```
{
  "jobConfiguration":
    {
      "commands":
        [{
          "type": "DEMAND_INTEGRATION_PERIOD",
          "active": "True"
        }]
    },
  "jobName": "Unique_Job_Name",
  "meters":
    [
      "ZR1J00005"
    ],
  "startJob": true,
  "timeoutTime": "2023-06-15T18:00:00+05:30",
  "jobType": "METER_COMMAND_GET",
  "callback":
    {
      "callbackOn":
        [
          "JOB_METER_CALLBACK",
          "JOB_CALLBACK"
        ],
      "callbackUrl": "MDMCallBackURL"
    }
}
```

-----Callback Response-----



**DIP GET Call Back
Response.json**

6.5 Demand Integration Period (SET)

-----API Request-----

```
{ "jobConfiguration":
  { "commands":
    [ {
      "type": "DEMAND_INTEGRATION_PERIOD",
      "activationTime": "2023-11-22T00:00:00+0530",
      "demandPeriod": 900
    }
  ],
  "jobName": "Unique_Job_Name",
  "meters":
  ["METER_D1_000394"],
  "startJob": true,
  "timeoutTime": "2023-06-15T18:00:00+05:30",
  "jobType": "METER_COMMAND_SET",
  "callback":
  { "callbackOn":
    [
      "JOB_METER_CALLBACK",
      "JOB_CALLBACK"
    ],
    "callbackUrl": "MDMCallBackURL"
  }
}
```

-----Callback Response-----



**DIP SET Call Back
Response.json**

6.6 Profile Capture Period (GET)

Field	Type	Default	Definition
profileType	MeterProfileType	REQUIRED	The profile on which to set the capture period. {Refer section 2}
capturePeriod	Period	REQUIRED	The required capture period in seconds. Period: defined in milli-seconds (ms)

```

-----API Request-----
{
  "jobConfiguration": {
    "commands": [
      {
        "type": "PROFILE_CAPTURE_PERIOD",
        "active": "True"
      }
    ]
  },
  "jobName": "Unique_Job_Name",
  "meters": [
    "ZR1J00006"
  ],
  "startJob": true,
  "timeoutTime": "2024-06-24T13:30:00+05:30",
  "jobType": "METER_COMMAND_GET",
  "callback": {
    "callbackOn": [
      "JOB_METER_CALLBACK",
      "JOB_CALLBACK"
    ],
    "callbackUrl": "MDMCallBackURL"
  }
}

-----Callback Response-----

```



PCP GET Call Back
Response.json

6.7 Profile Capture Period (SET)

```

-----API Request-----
{
  "jobConfiguration":
  {
    "commands":
      [ {
        "type": "PROFILE_CAPTURE_PERIOD",
        "profileType": "BLOCK_LOAD",
        "capturePeriod": 900
      } ],
    "jobName": "Unique_Job_Name",
    "meters": [
      "ZR1J00005"
    ],
    "startJob": true,
    "timeoutTime": "2024-06-24T12:50:00+05:30",
    "jobType": "METER_COMMAND_SET",
    "callback": {
      "callbackOn": [
        "JOB_METER_CALLBACK",
        "JOB_CALLBACK"
      ],
      "callbackUrl": "MDMCallBackURL"
    }
  }
}

-----Callback Response-----

```



PCP SET Call Back
Response.json

6.8 Max. Demand Reset

```
-----API Request-----
{
  "jobConfiguration": {
    "commands": [
      {
        "type": "MAX_DEMAND_RESET"
      }
    ]
  },
  "jobName": "Unique_Job_Name",
  "meters": [
    "AME1003241"
  ],
  "startJob": true,
  "timeoutTime": "2024-06-24T15:10:00+05:30",
  "jobType": "METER_COMMAND_SET",
  "callback": {
    "callbackOn": [
      "JOB_METER_CALLBACK",
      "JOB_CALLBACK"
    ],
    "callbackUrl": "MDMCallBackURL"
  }
}

-----Callback Response-----
```



**Max Demand Reset
Job Call Back Response**

6.9 Billing Period (GET)

Field	Type	Default	Definition
Date	Date (Integer)	REQUIRED	The date on which to start a new billing period. Only "dayOfMonth" can be set from 1 to 28, it is not allowed to set for 29, 30 and 31
billingCycle	BillingCycle	OPTIONAL	The billing cycle configuration {Refer below table}

Element	Values	Definition
billingCycle	"MONTHLY"	Billing for every month. DEFAULT value if it is not configured.
	"ODD_MONTHLY"	Billing occurs every odd month, i.e. January, March, May, July, September and November.
	"EVEN_MONTHLY"	Billing occurs every even month, i.e. February, April, June, August, October and December.

-----API Request-----

```
{
  "jobConfiguration": {
    "commands": [
      {
        "type": "BILLING_PERIOD",
        "active": "True"
      }
    ],
    "jobName": "Unique_Job_Name",
    "meters": [
      "A5722914"
    ],
    "startJob": true,
    "timeoutTime": "2024-02-15T13:45:00+05:30",
    "jobType": "METER_COMMAND_GET",
    "callback": {
      "callbackOn": [
        "JOB_METER_CALLBACK",
        "JOB_CALLBACK"
      ]
    }
  },
  "callbackUrl": "MDMCallBackURL"
}
```

```
}
}
```

-----Callback Response-----



BP GET Job Call Back
Response.json

6.10 Billing Profile (SET)

-----API Request-----

```
{
  "jobConfiguration": {
    "commands": [
      {
        "type": "BILLING_PERIOD",
        "date": {
          "dayOfMonth": 10
        }
      }
    ],
  },
  "jobName": "Unique_Job_Name",
  "meters": [
    "METER_D1_000394"
  ],
  "startJob": true,
  "timeoutTime": "2023-06-15T18:00:00+05:30",
  "jobType": "METER_COMMAND_SET",
  "callback": {
    "callbackOn": [
      "JOB_METER_CALLBACK",
      "JOB_CALLBACK"
    ],
    "callbackUrl": "MDMCallBackURL"
  }
}
```

-----Callback Response-----



**BP SET Job Call Back
Response.json**

6.11 Activity Calendar (GET)

Structure defining the calendar used as part of the meters time of use configuration.

Field	Type	Default	Definition
name	String	OPTIONAL	Unique identifier for this time of use calendar.
seasons	TimeOfUseSeason[]	REQUIRED	The season profiles defined in this calendar. {Refer section 6.12.1}
weeks	TimeOfUseWeek[]	REQUIRED	The week profiles defined in this calendar. {Refer section 6.12.2}
days	TimeOfUseDay[]	REQUIRED	The day profiles defined in this calendar. weeks and specialDay refer to this array. The first element of this array is index 0. {Refer section 6.12.3}
specialDay	SpecialDay[]	OPTIONAL	The special days defined in this calendar. {Refer section 6.12.4}

6.11.1 Time of Use Season

Field	Type	Default	Definition
name	String	OPTIONAL	Unique identifier for this season profile.
start	Date	REQUIRED	The date on which this season begins. {Refer section 6.12.6}
week	Integer	REQUIRED	The index into the calendars list of week profiles.

6.11.2 Time of Use Week

Field	Type	Default	Definition
days	Integer[7]	REQUIRED	Defines the profile used for each day of the week, based on the index into the list of day profiles. Monday to Sunday.

6.11.3 Time of Use Day

Field	Type	Default	Definition
Schedule	TimeOfUseAction[]	REQUIRED	The schedule for this day profile. {Refer section 6.12.4}

6.11.4 Time of Use Action

Field	Type	Default	Definition
time	Timestamp (HH:MM)	REQUIRED	The time at which the action should be executed.
zone	Integer	REQUIRED	The script action to execute. Script 1 – 16 tariff zones 1 to 16. Script 17 – 32 max demand zones 1 to 16.

6.11.5 Special Day

Field	Type	Default	Definition
dayOfMonth	Integer	REQUIRED	The day of the month for the special day.
month	Integer	REQUIRED	The month of the special day.
year	Integer	OPTIONAL	The year for the special day. If not specified this will be treated as a wildcard implying any year.
dayId	Integer	REQUIRED	Index for the element in the “days” array with “schedule”.

6.11.6 Date

Field	Type	Default	Definition
year	Integer	OPTIONAL	The year for the date. If not specified this will be treated as a wildcard implying any year.
month	Integer	OPTIONAL	The month for the date. If not specified this will be treated as a wildcard implying any month.
dayOfMonth	Integer	OPTIONAL	The day of the month for the date. If not specified this will be treated as a wildcard implying any day of the month.
dayOfWeek	Integer	OPTIONAL	The day of the week. If not specified this will be treated as a wildcard implying any day of the week.

-----API Request-----

```
{
  "jobConfiguration": {
    "commands": [
      {
        "type": "TARIFF_CALENDAR",
        "active": true
      }
    ]
  },
  "jobType": "METER_COMMAND_GET",
  "jobName": "Unique_Job_Name",
  "meters": [
    "A5722914"
  ]
},
```

```
"startJob": true,
"timeoutTime": "2024-06-24T15:30:00+05:30",
"callback": {
  "callbackOn": [
    "JOB_METER_CALLBACK",
    "JOB_CALLBACK"
  ],
  "callbackUrl": "MDMCallBackURL"
}
```

-----Callback Response-----



**Tariff Calendar GET
Job Callback Response**

6.12 Activity Calendar/TOU/TOD (SET)

-----API Request-----

TOU SET for 3 Phase Meter in 4 Zones - Linkwell



**Linkwell 3P TOU SET
Request.json**

TOU SET for 3 Phase Meter in 4 Zones - HPL



**HPL 3P TOU SET
Request.json**

TOU SET for 1P/3P/LTCT Meter in 7 Zones – Genus



**Genus 1P TOU SET
Request.json**

TOU SET for 1P/3P Meter in 4 Zones – AEW



AEW TOU SET 1p and
3p.json

TOU SET for 1P/3P meter in 8 zones – Schneider



SEI TOU SET 1p and
3p.json

-----Callback Response-----



Tariff Calendar SET
Job Callback Respons

6.13 Load Curtailment (GET)

Field	Type	Default	Definition
loadCurtailmentState	LoadCurtailmentState	OPTIONAL	Define the operating state of the load curtailment within the meter.
powerLimitNormal	ActivePower	OPTIONAL	The normal mode, power limit trip point.
powerLimitEmergency	ActivePower	OPTIONAL	The emergency mode, power limit trip point.
powerLimitMinOverThresholdDuration	Period	OPTIONAL	The minimum period above the active threshold before performing the over threshold action.
powerLimitMinUnderThresholdDuration	Period	OPTIONAL	The minimum period below the active threshold before performing the under-threshold action.
currentLimitNormal	ActiveCurrent	OPTIONAL	The normal mode, current limit trip point.
currentLimitEmergency	ActiveCurrent	OPTIONAL	The emergency mode, current limit trip point.
currentLimitMinOverThresholdDuration	Period	OPTIONAL	The minimum period above the active threshold before performing the over threshold action.
currentLimitMinUnderThresholdDuration	Period	OPTIONAL	The minimum period below the active threshold before performing the under-threshold action.

alertPeriod	Period	OPTIONAL	The period before generating a load curtailment alert.
lockoutPeriod	Period	OPTIONAL	The period between relay lockout events.
lockoutMaxCounter	Integer	OPTIONAL	The maximum number of relay lockouts.

-----API Request-----

```
{ "jobConfiguration": {
  "commands": [
    { "type": "LOAD_CURTAILMENT", "active": true}
  ]
},
"jobType": "METER_COMMAND_GET",
"jobName": "Unique_Job_Name",
"meters": ["AME1003241"],
"startJob": true,
"timeoutTime": "2024-06-24T15:30:00+05:30",
"callback": { "callbackOn": [
  "JOB_METER_CALLBACK",
  "JOB_CALLBACK"
],
"callbackUrl": "MDMCallBackURL"
}
}
```

-----Callback Response-----


Load Curtailment
GET Job Callback Resp

6.14 Load Curtailment (SET)

Load Curtailment State - Enumerated type defining the allowed modes of operation for load curtailment, represented as a JSON string.

Value	Definition

ENABLED	Enable load curtailment.
DISABLED	Disable load curtailment.

```

-----API Request-----
{
  "jobConfiguration": {
    "commands": [
      {
        "type": "LOAD_CURTAILMENT",
        "loadCurtailmentState": "ENABLED",
        "powerLimitNormal": 100000,
        "powerLimitMinOverThresholdDuration": 60000,
        "powerLimitMinUnderThresholdDuration": 60000,
        "alertPeriod": 60000,
        "lockoutPeriod": 60000,
        "lockoutMaxCounter": 10
      }
    ]
  },
  "jobName": "Unique_Job_Name",
  "meters": [
    "METER_D1_000394"
  ],
  "startJob": true,
  "timeoutTime": "2023-06-15T18:00:00+05:30",
  "jobType": "METER_COMMAND_SET",
  "callback": {
    "callbackOn": [
      "JOB_METER_CALLBACK",
      "JOB_CALLBACK"
    ],
    "callbackUrl": "MDMCallBackURL"
  }
}

```

-----Callback Response-----



Load Curtailment SET
Job Callback Respons

6.15 Enable & Disable Net Metering

6.15.1 GET Meter Mode

```

-----API Request-----
{
  "jobConfiguration": {

```

```

"commands": [{
  "type": "METERING_MODE",
  "active": "True"
}],
"jobType": "METER_COMMAND_GET",
"jobName": "Unique_Job_Name",
"meters": [
  "A6207017"
],
"startJob": "true",
"timeoutTime": "2024-06-24T16:57:00+05:30",
"callback": {
  "callbackOn": [
    "JOB_METER_CALLBACK",
    "JOB_CALLBACK"
  ],
  "callbackUrl": "MDMCallBackURL"
}
}

```

----- Callback Response -----



**MM GET Job Callback
Response.json**

6.15.2 SET Import-export Mode (Net Metering Mode)

```

----- API Request -----
{
  "jobConfiguration":
  {
    "commands":
    [{

```

```

        "type": "METERING_MODE",
        "meteringMode": "IMPORT_EXPORT"
    }
},
"jobType": "METER_COMMAND_SET",
"jobName": "Unique_Job_Name",
"meters":
[
    "A6207017"
],
"startJob": "true",
"timeoutTime": "2025-06-25T12:57:00+05:30",
"callback":
{
    "callbackOn":
    [
        "JOB_METER_CALLBACK",
        "JOB_CALLBACK"
    ],
    "callbackUrl": "MDMCallBackURL"
}
}

```

-----Callback Response-----



MM SET IE Job
Callback Response.js

6.15.3 SET-Import only Mode (Forward Metering Mode)

-----API Request-----

```

{
    "jobConfiguration":
    {
        "commands":
        [{
            "type": "METERING_MODE",
            "meteringMode": "IMPORT"
        }
    ],
    "jobType": "METER_COMMAND_SET",
    "jobName": "Unique_Job_Name",
    "meters":
    [
        "A6207017"
    ],
    "startJob": "true",
    "timeoutTime": "2025-06-25T12:57:00+05:30",
    "callback":
    {
        "callbackOn":
        [

```

```

        "JOB_METER_CALLBACK",
        "JOB_CALLBACK"
    ],
    "callbackUrl": "MDMCallBackURL"
}
}

```

-----Callback Response-----



MM SET IMPORT Job
Callback Response.js

6.16 Enable & Disable Prepaid

6.16.1 Payment Mode

Enumerated type defining the payment mode to configure.

Value	Definition
PREPAID	Account is required to have funds credited prior to usage. Meter will be disconnected when the funds are depleted.
POSTPAID	Account is paid retrospectively based on measured usage. Meter may be disconnected if bill is not paid according to contract terms.

Field	Type	Default	Definition
mode	PaymentMode	OPTIONAL	The payment mode to be used by the meter. {Refer above table}
token	PaymentToken	OPTIONAL	Details of the last token update. {Refer section 6.18.1}
balance	PaymentBalance	OPTIONAL	The current balance information. {Refer section 6.18.2}

6.16.2 GET Payment Mode

-----API Request-----

```

{
  "jobConfiguration":
  {
    "commands":
    [
      {
        "type": "PAYMENT",
        "active": "True"
      }
    ],
    "jobType": "METER_COMMAND_GET",
    "jobName": "Unique_Job_Name",
    "meters":

```

```
[
  "A6207017"
],
"startJob": "true",
"timeoutTime": "2024-06-24T17:55:00+05:30",
"callback":
{
  "callbackOn":
  [
    "JOB_METER_CALLBACK",
    "JOB_CALLBACK"
  ],
  "callbackUrl": "MDMCallBackURL"
}
}
```

-----Callback Response-----



Payment Method GET
Job Callback response

6.16.3 Postpaid to prepaid

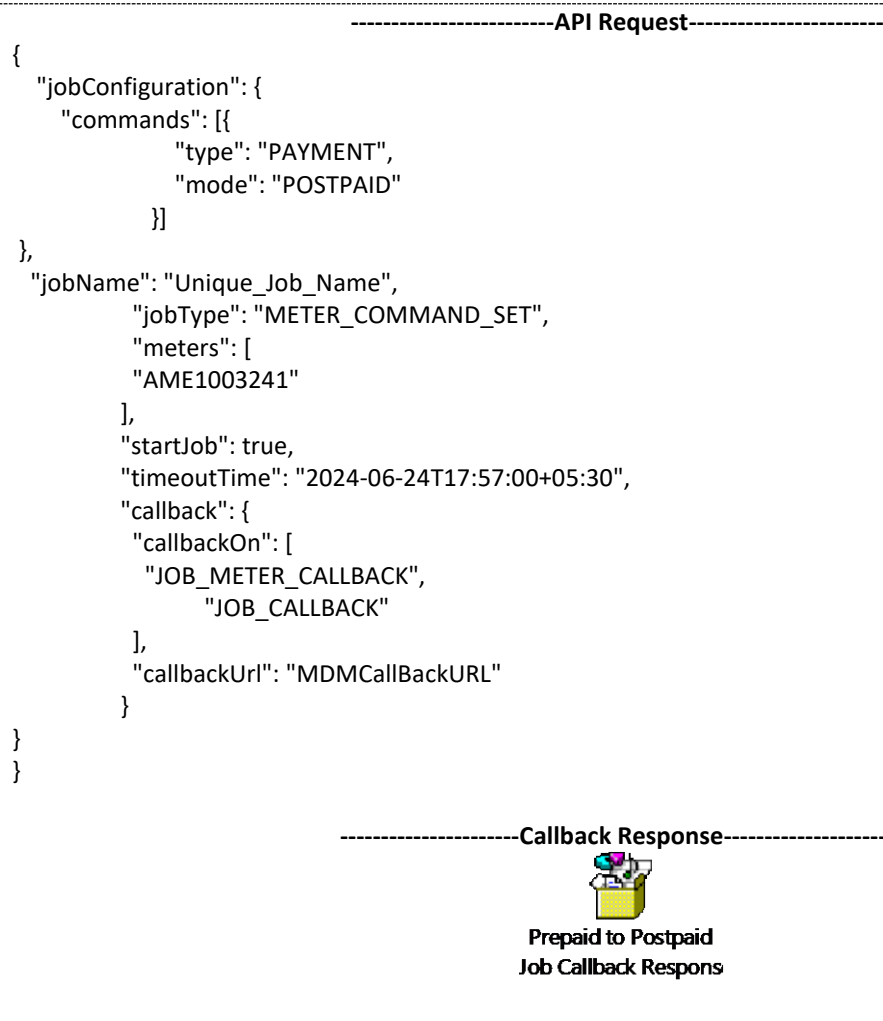
```
-----API Request-----
{
  "jobConfiguration": {
    "commands": [{
      "type": "PAYMENT",
      "mode": "PREPAID"
    }]
  },
  "jobName": "Unique_Job_Name",
  "jobType": "METER_COMMAND_SET",
  "meters": [
    "AME1003241"
  ],
  "startJob": true,
  "timeoutTime": "2024-06-24T17:57:00+05:30",
  "callback": {
    "callbackOn": [
      "JOB_METER_CALLBACK",
      "JOB_CALLBACK"
    ],
    "callbackUrl": "MDMCallBackURL"
  }
}
}
```

-----Callback Response-----



Postpaid to Prepaid
Job Callback Respons

6.16.4 Prepaid to postpaid



6.17 Recharge Balance Display on Meter screen (SET)

Individual Parameter SET is not suggested to be done as part of usual business case; functionality is supported by HES. For reference giving all the parameters required to be updated while doing a recharge.

6.17.1 Payment Token

Field	Type	Default	Definition
-------	------	---------	------------

Amount	Currency	REQUIRED	The value of the token, In India the units used would be paisa, or 0.01 INR.
Time	Date time stamp (yyyy-mm-ddTHH:MM:SS+5:30) For Ex.: 2024-08-22T13:50:00+05:30	REQUIRED	Time at which the token was applied.
amountAtLastRecharge	Currency	OPTIONAL	The account balance at the time the token was applied. In India the units used would be paisa, or 0.01 INR.

6.17.2 Payment Balance

Field	Type	Default	Definition
amount	Currency	REQUIRED	The current account balance. In India the units used would be paisa, or 0.01 INR.
time	Date time stamp (yyyy-mm-ddTHH:MM:SS+5:30) For Ex.: 2024-08-22T13:50:00+05:30	REQUIRED	Time at which this balance was correct.

```

-----API Request-----
{
  "jobConfiguration":
  {
    "commands":
    [{
      "type": "PAYMENT",
      "mode": "PREPAID",
      "token":
      {
        "amount": 100,
        "time": "2023-09-08T12:40:00+0530",
        "amountAtLastRecharge": 20
      }
    },
    {
      "balance":
      {
        "amount": 120,
        "time": "2023-09-08T12:40:00+0530"
      }
    }
  ],
  "jobName": "Unique_Job_Name",
  "jobType": "METER_COMMAND_SET",
  "meters":
  [
    "AME1003241"
  ],
  "startJob": true,
  "timeoutTime": "2024-06-24T15:47:00+05:30",

```

```

"callback":
{
    "callbackOn":
    [
        "JOB_METER_CALLBACK",
        "JOB_CALLBACK"
    ],
    "callbackUrl": "MDMCallBackURL"
}
}

```

-----Callback Response-----



**Recharge Display on
Meter SET Job Callbac**

6.18 Last Token Recharge Amount (GET)

-----API Request-----

```

{
  "jobConfiguration": {
    "commands": [{
      "type": "PAYMENT",
      "active": "True"
    }]
  },
  "jobType": "METER_COMMAND_GET",
  "jobName": "Unique_Job_Name",
  "meters": [
    "AME1003241"
  ],
  "startJob": true,
  "timeoutTime": "2024-06-24T15:45:00+05:30",
  "callback": {
    "callbackOn": [
      "JOB_METER_CALLBACK",
      "JOB_CALLBACK"
    ],
    "callbackUrl": "MDMCallBackURL"
  }
}

```

-----Callback Response-----



**Last Token Recharge
Amount GET Job Callb**

6.19 Meter Firmware upgrade over the air (FOTA)

The ability to perform remote firmware upgrade is a key benefit of a modern AMI system. This allows bugs to be fixed, features to be delivered iterably throughout the deployment and new functionality to be added during the overall project lifetime. Method mentioned below provides a simple way to update the firmware of the meters using RESTful API.

6.19.1 Pre-requisites

Before we start with firmware upgrade process and specific guideline to follow. There are few key factors which are required as pre-requisites

1. Meter firmware binary file – Provided by meter manufacturer
2. Image identifier (firmwareIdent) – Image identifier is provided by meter manufacturer along with binary file and generally remains same for all meters in one project, but in case image identifiers differs with respect to meter then meter manufacturers should provide image identifiers with respect to each meter and in same way it should be used in API mentioned below using “firmwareIdent” key in “meters” array list.
3. List of Meters – values to be populated in “id” key in “meters” array list. As to have better SLAs, visibility and traceability it is always recommended to do one meter per job meter FOTA.
4. HEX value (firmwareImage) – Convert binary file provided by meter manufacturer into HEX value and populate this convert HEX value in “firmwareImage” key in “commands” array.
5. MD5 Hash Key (firmwareHash) – It is an important key factor in firmware upgrade Job. It needs to be generated using binary file. Knowing MD5 is the Open Standard algorithm, MDM can use any of the standard library to get MD5 Hash Key from binary file and same to be populated in “firmwareHash” key in “commands” array.
6. firmwareIdent in jobConfiguration – This value to be filled in with meter firmware version.
7. “delayBetweenBlocksSecs” should be set as 1.

Please note –

1. Prioritise meter list based on hop count of meters. Hop count information should be provided by CyanConnode Project Team.
2. Default timeout should be provided as 24hrs as per SAT SLA, but in usual scenarios timeout can be provided as after 7 days.
3. JSON format only accepts hex conversion of Bin file so MDM must make sure to convert it before using in Firmware upgrade Job.
4. Providing meter firmware and image identifier shall be part of business process between MDMS and meter manufacturer.

-----API Request-----

```
{
  "jobName": "Unique_Job_Name",
  "meters": [
    {
      "id": "MSN1234",
      "firmwareIdent": "ABCD1234" --- This is to be provided by meter
      manufacturer and in HEX.
    },
    {
      "id": "MSN1233",
      "firmwareIdent": "ABCD1234" --- This is to be provided by meter
      manufacturer and in HEX.
    }
  ],
  "jobType": "METER_FIRMWARE_UPGRADE",
  "jobConfiguration": {
    "commands": [
      {
        "type": "FIRMWARE_IMAGE",
        "firmwareImage": "HEX_ENCODED_FIRMWARE_DATA", ---
        Input HEX conversion of Bin file provided by meter manufacturer.
        "firmwareHash": "MD5_HASH_OF_FIRMWARE_DATA", ---
        Generate and input MD5 Hash key using same Bin file.
        "delayBetweenBlocksSecs": 1,
        "firmwareIdent": "FIRMWARE_VERSION" --- Firmware
        version to which meter will be upgraded.
      }
    ]
  },
  "startJob": true,
  "timeoutTime": "2026-06-24T15:45:00+05:30",
  "callback": {
    "callbackOn": [
      "JOB_METER_CALLBACK",
      "JOB_CALLBACK"
    ],
    "callbackUrl": "MDMCallBackURL"
  }
}
```

-----Callback Response-----



FOTA Job Callback
response.json

7 Manage Meter Job

API URL	DomainName/sma/ws/manageMeterJob/\$JobName
API Method	POST

- This API is used to Manage Jobs created, where Job can be cancelled or start or delete.
- “**\$JobName**” will be variable value and it will be simply Job name for which command need to be executed.
- Possible Values for Instruction are "**START**", "**CANCEL**", "**DELETE**".

Value	Definition
START	Job should be started.
CANCEL	Job should be cancelled, with no further delivery to be performed.
DELETE	Job, and all associated data, should be deleted from the system.

```
{
  "instruction": "START"
}
```

RESPONSNE CODE: 200

```
{
}
```

Response code

Code	Definition
200	Request Initiated
401	Unauthorized
403	Forbidden
404	Not Found

8 Query Meter Job

API URL	DomainName/sma/ws/queryMeterJob/\$JobName
API Method	GET

Response code

Code	Definition
401	Unauthorized
403	Forbidden
404	Not Found

- This API is used to Query Meter Job created, through which user can get status and stats of Job initiated.
- “\$JobName” will be variable value and it will be simply Job name for which command need to be executed.
- This API is built to test initial integration between HES and MDM, shouldn't be used in production environment without prior permission from CyanConnode SPOC.
- Other than Job name in URL, no need to pass any other parameter in API request to get response.

9 Meter Status for Job

API URL	DomainName/sma/ws/meterStatusForJob/\$JobName
API Method	POST

- This API is used to get status data with respect to each meter defined in meter Job initiated.
- “\$JobName” will be variable value and it will be simply Job name for which command need to be executed.
- This API is built to test initial integration between HES and MDM, shouldn't be used in production environment without prior permission from CyanConnode SPOC.

-----API Request-----
{
}

Response code

Code	Definition
201	Request Initiated

401	Unauthorized
403	Forbidden
404	Not Found

10 Search Meters

API URL	DomainName/sma/ws/searchMeters
API Method	POST

- This API is used to get meter raw data and sends back Sync response immediately.
- This API can work as a streaming API by defining parameters like **"itemCount"** and **"itemStart"**.
- Maximum value for **"itemCount"** parameter will be 10K for single run.

```

-----API Request-----
{
  "itemCount": 1,
  "itemStart": 1
}

-----Response-----
[
  {
    "meterId": "A5543957",
    "nodeId": "00-1b-c5-0c-60-2b-1c-11",
    "vendor": "Schneider Electric India Private Limited",
    "firmwareVersion": "TE01.00",
    "hardwareVersion": "Single phase a.c. static direct connected watt-hour smart",
    "createTime": "2024-06-05T12:22:18+05:30",
    "updateTime": "2024-06-07T18:34:31+05:30"
  }
]

```

```

[
  {
    "meterId": "A5543957",
    "nodeId": "00-1b-c5-0c-60-2b-1c-11",
    "vendor": "Schneider Electric India Private Limited",
    "firmwareVersion": "TE01.00",
    "hardwareVersion": "Single phase a.c. static direct connected watt-hour smart",
    "createTime": "2024-06-05T12:22:18+05:30",
    "updateTime": "2024-06-07T18:34:31+05:30"
  }
]

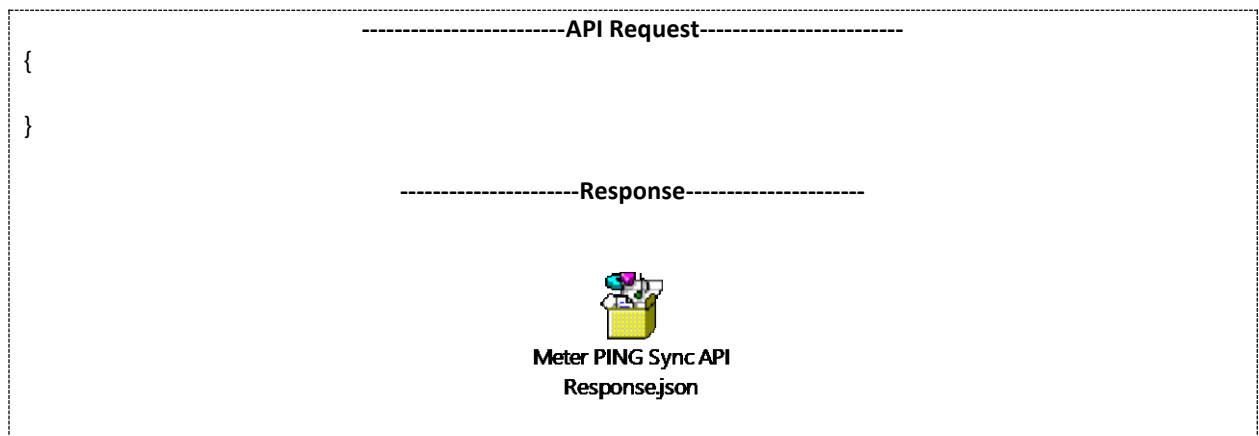
```

Response code

Code	Definition
201	Request Initiated
401	Unauthorized
403	Forbidden
404	Not Found

11 Meter PING API – Sync Response

API URL	DomainName/sma/ws/meterPing/\$meterID
API Method	GET



Response code

Code	Definition
200	Success
401	Unauthorized
403	Forbidden
404	Not Found

12 To Set Meter longitude-latitude

Below steps must be followed

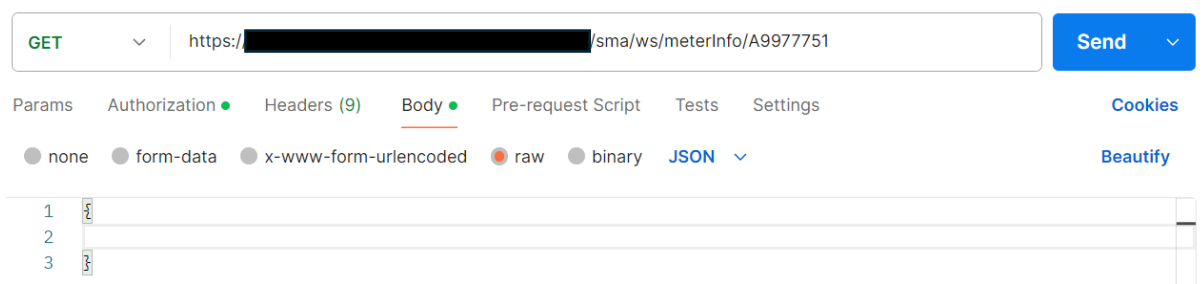
13.1. Using Meter serial number or Device Id, get Node Id using below API

API URL	DomainName/sma/ws/meterInfo/\$meterserialnumber
API Method	GET
Type of API	REST

Authentication	Basic Auth
----------------	------------

\$meterserialnumber – this is variable where need to pass meter serial number or device ID for which need to get Node ID.

Input



Output response

```
{
  "meterId": "A9977751",
  "nodeId": "00-1b-c5-0c-60-70-79-e1",
  "vendor": "Schneider Electric India Private Limited",
  "firmwareVersion": "014400CBO0430000",
  "hardwareVersion": "Three phase a.c. static direct connected watt-hour smart",
  "createTime": "2025-08-09T15:20:41+05:30",
  "updateTime": "2026-03-17T18:30:31+05:30"
}
```

13.2. Then using Node ID and below SOAP API longitude-latitude can be set.

API URL/WSDL	HESURL/nms/ws/NodeServices
API Method	----
Type of API	SOAP
Authentication	Basic Auth

-----Input XML-----

```
<soapenv:Envelope
xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:ws="http://ws.terminal.service.connode.com/">

  <soapenv:Header/>

  <soapenv:Body>

    <ws:setLocation>

      <nodeId></nodeId>

      <latitude></latitude>

      <longitude></longitude>

    </ws:setLocation>

  </soapenv:Body>

</soapenv:Envelope>
```